



Week 4 Lecture 1

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	23
▼ Type	Lecture
☰ Week #	4

Persistent Storage

Example: Gradebook

- Students: ID, name, address, ...
- Courses: ID, name, department, year, ...
- Student-Course Relationship: which students are registered for which courses

Gradebook

Aa Name	☰ IDNumber
<u>Sunil Shashi</u>	MAD001
<u>Chetana Anantha</u>	MAD002
<u>Madhur Prakash</u>	MAD003
<u>Nihal Surya</u>	MAD004
<u>Shweta Lalita</u>	MAD005
<u>Raghu Balwinder</u>	MAD006
<u>Gulshan Kuldeep</u>	MAD007
<u>Kishan Shrivatsa</u>	MAD008
<u>Purnima Sunil</u>	MAD009
<u>Nikitha Madhavi</u>	MAD010
<u>Lilavati Prabhakar</u>	MAD011
<u>Rama Yamuna</u>	MAD012

Gradebook

Aa CourseID	☰ Name
<u>EE1001</u>	Introduction to Electrical Engineering
<u>AM1100</u>	Engineering Mechanics
<u>MA1020</u>	Functions of Several Variables
<u>ME1100</u>	Thermodynamics
<u>BR1010</u>	Life Sciences

Spreadsheets

- Arbitrary data organized into rows and columns
- Operations defined on cells or ranges
- Multiple inter-linked sheets within single spreadsheet

Any kind of tabular data - expressed in tables

Relationships

- Students - Course?
- Separate entry with full details - student name, ID, address, course ID, name, department, etc.
 - Redundant
- Separate table "joining" students with courses
 - Only ID specified
 - Relation specified with "Keys"

Aa A	≡ B	# C
<u>MAD001</u>	BT1010	78
<u>MAD002</u>	EE1001	30
<u>MAD005</u>	EE1001	68
<u>MAD009</u>	AM1100	62
<u>MAD012</u>	AM1100	77
<u>MAD007</u>	BT1010	41

Questions

- How should the underlying data should be stored?
 - Can it be made persistent - survive server restart?
- How should the relations be represented?
- Structured ways to represent, manipulate data?